

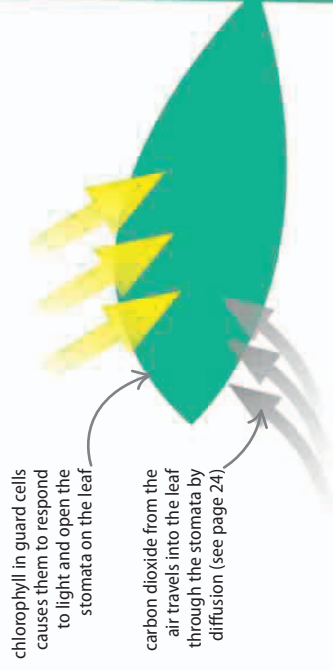
Photosynthesis

PLANTS MAKE THEIR OWN FOOD FROM SIMPLE INGREDIENTS AND SUNLIGHT.

Plants need sunlight to survive. They harness the energy in light to make food from carbon dioxide and water in a process called photosynthesis.

Light reaction

Photosynthesis is a chemical reaction that combines carbon dioxide gas and water to make a molecule of glucose. The glucose is the plant's food, and is sent around the plant to provide the energy it needs. The waste product of the process is oxygen. Photosynthesis itself is powered by sunlight. A chemical called chlorophyll in the leaves absorbs some of the light's energy and uses it to start the reaction.



△ Atmospheric carbon

During photosynthesis, carbon atoms are taken from the atmosphere. These atoms are the building blocks of all organic (carbon-containing) compounds—in both plants and the animals that eat them.

Leaf

A leaf is a plant's solar panel. It is flattened to create a larger surface area to catch as much sunlight as possible. The light shines through the surface of the leaf, and photosynthesis occurs in the cells inside. Water arrives from the plant along a vessel that runs down the center of the leaf. Carbon dioxide comes into the plant from the surrounding air through pores called stomata on the underside of the leaf.

Chloroplast

A green structure inside the cell where the chlorophyll is located.

Palisade cells

These column-shaped cells under the upper surface are where most of the photosynthesis takes place.

Vascular bundle

Xylem (blue) brings water and dissolved minerals to the leaf. Phloem (orange) takes away sucrose (see page 37).

Lower epidermis

The underside of the leaf is filled with pores called stomata (singular: stoma) that let gases in and out of the plant.

Upper epidermis

A layer of cells that forms the leaf's upper surface. These cells have a waxy coating to reduce the amount of water lost through evaporation.

Spongy mesophyll

Cells with large spaces between them where the gases circulate.

Water loss

Leaves lose water through evaporation and need a constant supply so they do not dry out.

Guard cells

A stoma is made of two guard cells, which move away from each other to open the pore when the Sun is shining, and move together to close it when it's dark.

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